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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
Third Semester, November, 2023

Paper Code: ECC-207**Subject: Digital Logic and Computer Design (CSE/IT/ECE/CST/ITE)****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Simplify the Boolean expression: $Y = A'B'C' + A'BC' + AB'C' + ABC'$. [Note: Prime notation denotes logical negation / inversion, e.g. $A' = \bar{A}$].	2	CO1
(b)	Convert: (i) $[110101]_{\text{Gray}} = [\quad ? \quad]_2$ (ii) $[643]_{10} = [\quad ? \quad]_{\text{Excess-3}}$.	2	CO1
(c)	What is a demultiplexer? Explain the difference between a demultiplexer and decoder.	2	CO1
(d)	What advantage does a J-K flip-flop have over an S-R flip-flop?	2	CO2
(e)	What is a shift register? Name the different types of shift registers.	2	CO2
2. (a)	Explain how a full subtractor is designed.	5	CO1
(b)	Realise J-K flip-flop using S-R flip-flop.	5	CO2
3. (a)	Simplify the expression $Y = \sum m(3, 4, 5, 7, 9, 13, 14, 15)$ using the K-map method.	5	CO1
(b)	Explain the operation of Master-slave J-K flip-flop and show how the race-around condition is eliminated in it.	5	CO2
4. (a)	Explain briefly the BCD-to-seven segment decoder.	5	CO1
(b)	Design a MOD-6 ripple counter using J-K flip-flops.	5	CO2

****** End of Question Paper ******